Question	Answer	Marks
4(a)(i)	temperature = –273.15 °C	**A1**
4(a)(ii)	temperature = 0 K	**A1**
4(b)(i)	gas is ideal	**B1**
4(b)(ii)	$pV = NkT$	**C1**
	$N = 270 / (8.0 \times 10^{-21})$ $= 3.4 \times 10^{22}$	**A1**
4(b)(iii)	$n = (3.4 \times 10^{22}) / (6.02 \times 10^{23})$ $= 0.056$ mol	**A1**
4(c)	$\frac{1}{2} m\langle c^2\rangle = (3 / 2)\, kT$	**C1**
	$\frac{1}{2} \times m \times 1900^2 = 1.5 \times 8.0 \times 10^{-21}$ ($m = 6.65 \times 10^{-27}$ kg)	**C1**
	$m = (6.65 \times 10^{-27}) / (1.66 \times 10^{-27})$	**C1**
	$= 4.0$ u	**A1**

Question	Answer	Marks